

Grade 6 Mathematics

Մաթեմատիկա — 6-րդ դասարան

COURSE CODE	MTH-G6
PROGRAM	Middle School Mathematics · Միջին դպրոցի մաթեմատիկա
LEVEL	Grade 6 · typical age 11-12
DURATION	84-96 instructional hours (approx. 72 lessons)
PREREQUISITE	Grade 5 Mathematics or equivalent: fraction and decimal operations, coordinate plane, volume.
LEADS TO	Grade 7 Mathematics (MTH-G7)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 6 is where arithmetic becomes mathematics. Three genuinely new ideas arrive: ratio and rate as a way of comparing quantities multiplicatively, negative numbers as an extension of the number line in both directions, and the variable as a symbol that can stand for any number. Students complete division of fractions, become fluent with all decimal operations, meet statistical variability for the first time, and begin to write and solve one-step equations. The habits formed this year determine whether algebra will feel like a natural continuation or like a foreign language.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Վարժարան treats Grade 6 as the beginning of the algebra pathway, not the end of elementary school. Teachers are asked to insist on precise language — a ratio is not a fraction, an expression is not an equation, a variable is not an abbreviation — because imprecision that is tolerable in Grade 6 becomes disabling in Algebra I. Ratio reasoning is developed with tables, tape diagrams, and double number lines before any cross-multiplication is permitted; students who reach for cross-multiplication early consistently fail the non-routine proportional problems of Grade 7.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Performs all four operations with fractions and decimals accurately	The computational floor for the entire course; weakness here shows up as ratio errors, not arithmetic errors.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Explains why dividing by $\frac{1}{2}$ gives a larger result	Signals whether fraction division was understood or memorized; predicts success in Unit 1.
Graphs points in the first quadrant of the coordinate plane	Prerequisite for Unit 3 and for all later graphing.
Evaluates an expression with parentheses using order of operations	Prerequisite for Unit 4; a very common gap in transferring students.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Understand ratio concepts and use ratio reasoning to solve real-world problems.
- 2 Find unit rates and use them to compare, convert, and solve rate problems.
- 3 Solve percent problems as rate-per-hundred, finding the part, the whole, or the percent.
- 4 Interpret and compute quotients of fractions and explain the meaning of the result.
- 5 Fluently divide multi-digit numbers and perform all four operations with multi-digit decimals.
- 6 Find greatest common factors and least common multiples and apply the distributive property.
- 7 Understand positive and negative numbers, absolute value, and ordering on the number line.
- 8 Graph points in all four quadrants and solve problems using coordinates and absolute value.
- 9 Write, read, and evaluate algebraic expressions and identify equivalent expressions.
- 10 Solve one-variable one-step equations and inequalities and represent solutions on a number line.
- 11 Analyze the relationship between dependent and independent variables using tables, graphs, and equations.
- 12 Find areas of triangles and quadrilaterals, volumes of prisms with fractional edges, and surface areas using nets.
- 13 Recognize a statistical question, summarize a distribution by center and spread, and interpret displays.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 10-11 LESSONS

Ratios and Rates

Հարաբերություններ և արագություններ

TOPICS COVERED

- The language of ratios: $a:b$, a to b , a/b , and when each is appropriate
- Part-to-part and part-to-whole ratios
- Equivalent ratios and ratio tables
- Tape diagrams and double number lines
- Unit rates and the meaning of 'per'
- Comparing rates to make a decision (better buy problems)
- Constant speed and distance-rate-time problems
- Converting measurement units by rate reasoning
- Plotting equivalent ratios on the coordinate plane
- Percent as a rate per hundred
- Finding the part, the whole, and the percent

LEARNING OBJECTIVES

- 1 Describe a ratio relationship in words and distinguish part-to-part from part-to-whole.
- 2 Generate equivalent ratios with a table and use the table to answer a question.
- 3 Compute a unit rate and use it to compare two options.
- 4 Solve a percent problem by treating percent as a rate per hundred, without a memorized formula.
- 5 Convert between measurement units by multiplying by a rate equal to one.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves ratio problems with a table or diagram in 8 of 10 trials.
- Computes and interprets a unit rate correctly in 9 of 10 trials.
- Solves all three percent question types (find part, whole, percent) at 85% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Ratios are treated additively: if 2:3 then 4:5.	The defining error of Grade 6. Build a ratio table and a tape diagram together; the diagram makes the additive answer visibly wrong.
Percent is treated as a whole number rather than a rate, so 25% of 60 is answered as 25.	Return to 'per hundred'. Use a 10×10 grid until the student can shade any percent and read the corresponding amount.

MISCONCEPTION	RECOMMENDED RESPONSE
Cross-multiplication is used before the ratio relationship is understood.	Withhold cross-multiplication entirely in this unit. It is a shortcut that hides the structure, and students who acquire it early cannot solve unfamiliar problems in Grade 7.

TEACHING NOTE

Ratio reasoning is the conceptual centre of middle school mathematics and the strongest predictor of Algebra I performance. Give this unit more time than the calendar suggests, and use visual models in every single lesson.

UNIT 2 • 9-10 LESSONS**The Number System: Fractions, Decimals, Factors**

Թվային համակարգ. կոտորակներ, տասնորդականներ, բաժանարարներ

TOPICS COVERED

- Dividing a fraction by a fraction with visual models
- The invert-and-multiply rule and why it works
- Word problems requiring fraction division
- Fluent multi-digit long division
- All four operations with multi-digit decimals
- Greatest common factor of two numbers to 100
- Least common multiple of two numbers to 12
- Factoring out a common factor with the distributive property
- Prime factorization

LEARNING OBJECTIVES

- 1 Divide a fraction by a fraction using a model and explain the result in context.
- 2 Justify the invert-and-multiply rule rather than merely applying it.
- 3 Compute the GCF and LCM of a pair of numbers and choose which is needed in a given problem.
- 4 Rewrite a sum such as $36 + 8$ as $4(9 + 2)$ using the distributive property.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Divides fractions accurately and interprets the quotient in context in 8 of 10 trials.
- Performs all four decimal operations at 90% accuracy.
- Selects GCF vs. LCM correctly in 8 of 10 word problems.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Invert-and-multiply is applied to the wrong fraction, or to multiplication problems.	Derive the rule once with a model: $3/4 \div 1/8$ asks how many eighths fit in three fourths. A student who has seen the derivation applies it correctly.

MISCONCEPTION	RECOMMENDED RESPONSE
GCF and LCM are confused in word problems.	Ask a diagnostic question: is the answer smaller than the numbers (splitting into groups → GCF) or larger (events lining up → LCM)?

TEACHING NOTE

The distributive-property factoring in this unit is quietly pre-algebra. Write $4(9 + 2)$ and then $4(x + 2)$ side by side in the final lesson to make the connection visible.

UNIT 3 • 8-9 LESSONS**Rational Numbers and the Coordinate Plane**

Ռացիոնալ թվեր և կոորդինատային հարթություն

TOPICS COVERED

- Negative numbers in real contexts: temperature, elevation, debt
- The number line extended in both directions
- Opposites and the meaning of $-(-a)$
- Ordering rational numbers and interpreting the order in context
- Absolute value as distance from zero
- Absolute value vs. order: $-7 < -3$ but $|-7| > |-3|$
- All four quadrants of the coordinate plane
- Reflections across the axes and sign changes
- Distance between points sharing a coordinate
- Polygons on the coordinate plane

LEARNING OBJECTIVES

- 1 Place any rational number on the number line and justify its position.
- 2 Interpret an inequality statement about negative numbers in a real context.
- 3 Explain absolute value as distance and distinguish it from ordering.
- 4 Find the distance between two points that share an x- or y-coordinate.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Orders rational numbers including negatives correctly in 9 of 10 trials.
- Distinguishes magnitude from order in 8 of 10 trials with justification.
- Graphs and reflects points across both axes accurately in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
-7 is judged greater than -3 because $7 > 3$.	The number line, always. Anchor with temperature: -7° is colder, therefore less. Insist on the context whenever the abstract ordering is unstable.
Absolute value is described as 'making a number positive'.	This survives until it fails on $ x = 5$ in Algebra. Define it as distance from zero from the first lesson and never use the shortcut phrase.

TEACHING NOTE

Negative numbers are the first idea in the curriculum with no direct physical model for the operations. Grade 6 builds only meaning and ordering; Grade 7 builds the operations. Do not rush ahead.

UNIT 4 • 10-11 LESSONS**Expressions and Equations**

Արտահայտություններ և հավասարումներ

TOPICS COVERED

- Exponents and evaluating numerical expressions with exponents
- Translating verbal phrases into algebraic expressions
- Vocabulary: term, coefficient, factor, constant, sum, product
- Evaluating expressions at given values
- Applying properties to generate equivalent expressions
- Combining like terms; expanding and factoring
- Identifying equivalent expressions by reasoning
- The meaning of a solution to an equation
- Solving one-step equations by inverse operations
- Writing an equation to represent a situation
- Inequalities of the form $x > c$ and $x < c$
- Representing inequality solutions on a number line

LEARNING OBJECTIVES

- 1 Translate a verbal description into an algebraic expression and back.
- 2 Determine whether two expressions are equivalent and justify with a property.
- 3 Solve a one-step equation and check the solution by substitution.
- 4 Write and graph an inequality representing a real constraint, and explain that it has infinitely many solutions.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Translates between words and symbols correctly in 9 of 10 trials.
- Solves one-step equations at 95% accuracy with checking shown.
- Explains why an inequality has infinitely many solutions.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A variable is read as a label rather than a number (3s means '3 students').	The 'student-professor' error, and it persists into calculus if untreated. Insist that s is <i>the number of</i> students. Substitute values immediately to make the variable behave like a number.
$3x$ is read as 'three then x' rather than as multiplication.	Evaluate $3x$ at several values in the first lesson. Ambiguity here corrupts everything downstream.

MISCONCEPTION	RECOMMENDED RESPONSE
The equals sign is still read as 'compute the answer'.	Use balance-scale imagery. Present equations with operations on both sides even before they are solvable, and ask whether they are true.

TEACHING NOTE

Precision of language is the whole point of this unit. Require the student to say 'the coefficient is 3' rather than 'the number in front', every time. Vocabulary is not decoration in algebra; it is how thinking is organized.

UNIT 5 • 9-10 LESSONS**Dependent Relationships, Area, and Volume**

Կախյալ մեծություններ, մակերես և ծավալ

TOPICS COVERED

- Independent and dependent variables
- Representing a relationship with a table, a graph, and an equation
- Moving between the three representations
- Area of triangles by decomposition and by formula
- Area of parallelograms, trapezoids, and rhombi
- Area of composite and irregular polygons
- Polygon area on the coordinate plane
- Volume of right rectangular prisms with fractional edge lengths
- Nets of three-dimensional figures
- Surface area from a net

LEARNING OBJECTIVES

- 1 Identify the independent and dependent variable in a situation and write an equation relating them.
- 2 Produce a graph from a table and a table from an equation for the same relationship.
- 3 Derive the triangle area formula by relating it to a parallelogram.
- 4 Compute the surface area of a prism or pyramid from its net.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Moves correctly among table, graph, and equation in 8 of 10 tasks.
- Computes areas of triangles and quadrilaterals at 90% accuracy.
- Computes surface area from a net correctly in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The slant side of a triangle is used as the height.	The height must be perpendicular to the chosen base. Draw the altitude explicitly on every diagram, including obtuse triangles where it falls outside.

MISCONCEPTION	RECOMMENDED RESPONSE
Surface area and volume are confused.	Unfold the net physically. Surface area is what you paint; volume is what you fill. Require square units and cubic units respectively.

TEACHING NOTE

The three-representations idea introduced here is the organizing structure of all of Algebra I. Name it explicitly and return to it whenever a new relationship appears.

UNIT 6 • 8-9 LESSONS**Statistics and Distributions**

Վիճակագրություն և բաշխումներ

TOPICS COVERED

- What makes a question statistical
- Variability as the reason statistics exists
- Dot plots, histograms, and choosing bin width
- Describing a distribution: shape, center, spread
- Mean as a fair share and as a balance point
- Median and when it is preferable to the mean
- The effect of an outlier on mean and median
- Mean absolute deviation
- Interquartile range and the five-number summary
- Box plots and comparing two distributions
- Reporting a measure of center with the context and units

LEARNING OBJECTIVES

- 1 Distinguish a statistical question from a non-statistical one and explain why.
- 2 Construct a dot plot, histogram, and box plot from the same data set and say what each reveals.
- 3 Choose and justify an appropriate measure of center for a given distribution.
- 4 Describe how an outlier affects the mean but not the median, with a computed example.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Identifies statistical questions correctly in 9 of 10 trials.
- Constructs all three display types accurately from raw data.
- Justifies a choice of center with reference to the shape of the distribution.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The mean is used automatically regardless of skew or outliers.	Use a salary data set with one extreme value. Compute both measures and ask which better describes a typical person.

MISCONCEPTION	RECOMMENDED RESPONSE
A histogram is treated as a bar graph.	Bars touch because the horizontal axis is continuous. Have the student change the bin width and observe how the shape changes — bar graphs cannot do this.

TEACHING NOTE

Statistics is often taught last and rushed. It should not be: this unit introduces variability, which is the entire conceptual basis of AP Statistics and of quantitative reasoning generally.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Grade 5 fraction and decimal operations plus early ratio reasoning; produces the initial academic profile.
Unit Check	End of each unit · 35 min	Unit mastery criteria plus two extended reasoning items requiring written justification.
Fluency Probe	Every 3rd lesson · 5 min	Fraction and decimal computation; maintained all year to prevent decay.
Ratio Reasoning Diagnostic	During and after Unit 1	Non-routine ratio problems that cannot be solved by cross-multiplication; the key predictor of Grade 7 readiness.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Grade 7 placement and eligibility for the accelerated algebra pathway.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60 min	Full course in approximately 50–55 weeks.
Accelerated	2 lessons/week	Full course in approximately 28–32 weeks; recommended for students targeting Algebra I in Grade 8.
Compacted 6/7	2 lessons/week	Grade 6 and Grade 7 compressed across two years for students targeting Algebra I in Grade 7.
Intensive / Catch-Up	2–3 lessons/week	22–26 weeks with diagnostic-driven ordering.
Summer Program	3 lessons/week, 6 weeks	Ratios and rates intensive (Unit 1) — the highest-demand summer course in the middle school band.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Thirty to forty minutes per lesson: six to nine problems including two word problems, one cumulative review item, and one problem requiring a written explanation rather than a numerical answer. From Unit 4 onward every assignment includes at least one problem where the student must translate between words and symbols in both directions. Students are expected to attempt every problem and to mark — rather than skip — the ones they could not finish; an honest blank is more useful to the teacher than a guessed answer.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Uses ratio tables, tape diagrams, and double number lines to solve ratio problems.
- ◆ Computes and interprets unit rates and solves all three percent problem types.
- ◆ Divides fractions by fractions and explains why the rule works.
- ◆ Understands negative numbers, absolute value, and ordering on the number line.
- ◆ Graphs and reasons in all four quadrants of the coordinate plane.
- ◆ Writes, reads, and evaluates algebraic expressions with correct vocabulary.
- ◆ Solves one-step equations and inequalities and checks solutions.
- ◆ Moves fluently among tables, graphs, and equations for a relationship.
- ◆ Finds areas of polygons and surface areas from nets.
- ◆ Summarizes a distribution by center and spread and justifies the choice.

- ◆ Completed Grade 6 Mathematics — ready for Grade 7.

MATERIALS AND RESOURCES

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- Digital ratio tables, tape diagrams, and double number lines on the shared whiteboard.
- Two-color counters and an extended number line for the rational-numbers unit.
- Nets of prisms and pyramids, printable for physical folding.
- Graphing tool for statistical displays (dot plot, histogram, box plot).
- Վարժարան Grade 6 problem bank — approximately 1,300 items, including a 200-item non-routine ratio sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.